

# SUPERSTORE PROFITABILITY, DISCOUNTING, CUSTOMER INSIGHTS, AND SALES FORECASTING

*An analytical and controlling perspective on Sample Superstore retail data*

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| <b>Dataset</b>             | Kaggle Sample Superstore — 9,994 transaction-line records, January 2014–December 2017                                             |
| <b>Tools</b>               | Python (pandas, NumPy, matplotlib, statsmodels, scikit-learn) and Power BI                                                        |
| <b>Companion notebooks</b> | Profitability Analysis · Rolling Sales Forecast                                                                                   |
| <b>Dashboard</b>           | Executive Overview · Forecast · Profitability · Customers & Geography · Management Action Plan · Customer Profitability Deep Dive |

## Executive Summary

The business generates approximately \$2.30M in sales and \$286.1K in profit, equal to a 12.46% profit margin. Aggregate profitability, however, conceals material variation across product lines, discount tiers, cities, customer accounts, and time periods.

The analysis identifies five management-relevant findings:

1. Loss exposure is material. 18.7% of transaction lines have negative profit.
2. Aggregate product profitability and loss exposure are different measures. Tables, Bookcases, and Supplies have negative aggregate profit in the Power BI model. Separately, Binders, Tables, Machines, and Bookcases account for 72.5% of loss dollars when the analysis considers only loss-making transaction lines.
3. Discounting is associated with lower profitability. Average transaction-line profit is positive at the 20% discount tier but negative at the observed 30% tier. In a controlled OLS model, a 10-percentage-point increase in discount is associated with approximately \$25 lower transaction-line profit.
4. Revenue is not sufficient evidence of value creation. Philadelphia is a top-five sales city but has negative aggregate profit. November sales are high, but margins are lower than typical surrounding months.
5. Customer profit is broadly distributed. 155 of 793 customers have negative aggregate profit, while the top 10 customers account for 17.5% of total profit.

The sales forecast has a validation error of 19.0% MAPE on a Jul–Dec 2017 holdout. Its Jan–Jun 2018 base-case projection is approximately \$305K, with a validation-error-based planning range of approximately \$247K–\$363K. This range is suitable for directional planning, not a hard revenue commitment.

## 1. Business Questions

This project examines a US retail dataset from a controlling and business-analysis perspective. It asks:

1. Which product categories, sub-categories, regions, cities, and customers generate profit or conceal losses?
2. How is discounting associated with transaction-line profitability after accounting for observed category, region, and segment differences?
3. Which customer segment drives the October sales decline, and is the decline caused by order volume, average order value, or both?
4. What level and range of sales should management use for six-month planning?
5. Which concrete actions should be prioritized, and how should their impact be measured?

## 2. Data and Analytical Units

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The Kaggle Sample Superstore dataset contains 9,994 transaction-line records from January 2014 through December 2017. Variables include order and shipment dates, customer identifiers, product category and sub-category, region, city, customer segment, ship mode, sales, quantity, discount, and profit.

| Analytical unit  | Definition and use                                                                                          |
|------------------|-------------------------------------------------------------------------------------------------------------|
| Transaction line | Raw observation level; used for category, sub-category, discount-tier, and transaction-line profit analysis |
| Order            | Aggregated by unique Order ID; used for order count and AOV                                                 |
| Customer         | Aggregated by unique Customer ID; used for customer analysis                                                |
| Month            | Sales aggregated to monthly frequency; used for forecasting                                                 |

The distinction between transaction-line and aggregate measures is essential. A sub-category can contain many loss-making transaction lines but still have positive aggregate profit after profitable lines are included.

Data preparation steps:

- Convert Order Date and Ship Date to datetime variables.
- Check data types, missing values, duplicate records, and discount ranges.
- Create Shipping Days = Ship Date – Order Date.
- Use distinct Order ID values for order counts and average order value.
- Aggregate by Customer ID for customer-level profitability.
- Retain raw transaction lines for product, category, discount-tier, and line-profit diagnostics.

## 3. Profitability Findings

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### 3.1 Category performance

Technology and Office Supplies are the largest profit contributors. Furniture has comparable sales volume but a substantially weaker margin.

| Category        | Sales     | Profit    | Profit margin |
|-----------------|-----------|-----------|---------------|
| Technology      | \$836,154 | \$145,455 | 17.40%        |
| Office Supplies | \$719,047 | \$122,491 | 17.04%        |
| Furniture       | \$741,999 | \$18,451  | 2.49%         |

Within Furniture, Chairs and Furnishings are profitable, while Tables and Bookcases have negative aggregate profit. The latter two also carry higher average discounts than their profitable Furniture peers.

### 3.2 Discount tiers

The transaction-line correlation between Discount and Profit is  $r = -0.219$ . This is a moderate negative association, not a complete explanation of individual line profitability.

- At a 20% discount, mean transaction-line profit remains positive at approximately \$24.7.
- At a 30% discount, mean transaction-line profit becomes negative at approximately -\$45.7.
- Higher discount tiers generally show increasingly negative average profit, although sparse tiers must be interpreted with caution.

The 20%–30% range should be treated as an observed profitability risk zone, not as a universally causal or mechanically exact break-even threshold.

### 3.3 Two complementary loss measures

| Measure                                | Definition                                                                       | Main result                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Negative aggregate-profit sub-category | Sum all transaction-line profits within each sub-category                        | Tables, Bookcases, and Supplies have negative aggregate profit in the Power BI model |
| Loss-line concentration                | Sum Profit only among transaction lines with Profit < 0, grouped by sub-category | Binders, Tables, Machines, and Bookcases account for 72.5% of loss dollars           |

These results do not conflict. Binders and Machines can have positive aggregate profit while still generating substantial loss exposure through particular products, customers, or discount terms.

### 3.4 Regression analysis

The baseline OLS model estimates:

$$Profit_i = \beta_0 + \beta_1 \cdot Discount_i + Category_i + Region_i + Segment_i + \varepsilon_i$$

The regression uses HC3 heteroskedasticity-robust standard errors. The estimated discount coefficient is -249.64 with  $p < 0.001$ . Conditional on the included category, region, and segment controls, a 10-percentage-point higher discount is associated with approximately \$25 lower profit per transaction line.

The model's  $R^2$  is approximately 5.7%. The purpose is diagnostic: it validates the direction and conditional magnitude of the discount association. It is not designed to predict individual transaction profit accurately.

The coefficient should not be interpreted causally. The data do not include COGS, product cost, promotion type, inventory pressure, marketing spend, negotiated terms, or other factors that may jointly affect discount and profit.

Two further results refine the category and regional findings above. Once discount and region are controlled for, Technology retains significantly higher line profit than the reference category, Furniture (+\$59.56,  $p < 0.001$ ), but Office Supplies is not statistically distinguishable from Furniture (+\$7.25,  $p = 0.207$ ), despite its materially healthier raw margin in Section 3.1. Regionally, West's per-line profit is not higher than Central's once discount and category are controlled for; if anything it is slightly lower (-\$16.18,  $p = 0.011$ ). This indicates West's higher total profit reflects greater order volume rather than a stronger per-order margin, a distinction the Price-Volume-Mix bridge below makes explicit. An interaction specification (Discount  $\times$  Category) tests whether the discount slope itself differs by category; results are reported in the companion notebook.

### 3.5 Price-Volume-Mix decomposition (2016 to 2017)

The regression above describes how discount relates to line profit but does not explain why total company profit changed year over year. A Price-Volume-Mix (PVM) bridge, computed at the category level in the companion notebook, splits the 2016-to-2017 profit change of approximately +\$11,644 into three components: a Volume Effect of approximately +\$16,650 from overall sales growth, a Mix Effect of approximately +\$3,746 from a shift toward higher-margin categories, and a Rate Effect of approximately -\$8,752 from declining margins within categories. The three effects sum exactly to the observed change; the notebook verifies this reconciliation with an assertion rather than assuming it holds.

The Rate Effect is concentrated in Furniture, whose margin fell from approximately 3.50% in 2016 to 1.40% in 2017, consistent with the discounting pattern in Sections 3.1 and 3.2. The management interpretation is that profit grew primarily because the business sold more, not because it became more efficient at converting sales into profit; continued margin erosion would offset future volume gains. This is the calculation behind the Profit Change Decomposition waterfall on the Profitability dashboard page.

## 4. Customers, Geography, and Seasonality

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### 4.1 Customer segments

Consumer is the largest and most profitable segment, generating approximately \$1.16M in sales and \$134,119

in profit. Corporate and Home Office generate approximately \$91,979 and \$60,299 in profit, respectively.

Average discounts are similar by segment: approximately 15.8% for Consumer, 15.8% for Corporate, and 14.7% for Home Office.

## 4.2 October Consumer dip

The October sales dip is concentrated in Consumer rather than Corporate or Home Office.

| Segment     | September sales | October sales | Change |
|-------------|-----------------|---------------|--------|
| Consumer    | \$185,056       | \$66,713      | –63.9% |
| Corporate   | \$74,020        | \$78,292      | +5.8%  |
| Home Office | \$48,574        | \$55,318      | +13.9% |

At the order level, Consumer order count falls from 351 to 197 (–43.9%), while Consumer average order value falls from approximately \$527 to \$339 (–35.8%). This identifies the mechanics of the decline, but does not establish an external cause such as promotions, marketing activity, competitors, or holiday timing.

## 4.3 November margin pressure

November is a strong sales month but not a correspondingly strong-margin month. Observed November margins range around 5%–8%, below the approximately 15% typical of surrounding months. This is consistent with a sales-volume/margin trade-off and warrants promotion-level testing.

## 4.4 Regional and city profitability

West and East are the strongest regions by aggregate profit. West produces approximately \$108,418 of profit and East approximately \$91,523. South produces approximately \$46,749, exceeding Central's approximately \$39,706.

Philadelphia is the critical exception to the assumption that high sales imply high value. It generates approximately \$109,077 in sales but –\$13,838 in profit. Product mix, discount rates, and customer composition should be compared with peer high-sales cities.

## 4.5 Customer profitability deep dive

Customer analysis covers 793 unique customers. 155 customers have negative aggregate profit. The top 10 customers account for 17.5% of total profit, indicating that total customer profit is not highly concentrated in a few accounts.

The customer sales-versus-profit chart identifies several high-sales, negative-profit accounts. These should be investigated using order frequency, average order value, product mix, discounting, and—where available—service cost.

## 5. Sales Forecasting

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### 5.1 Model design

Sales are aggregated monthly from January 2014 through December 2017, producing 48 monthly observations. The selected model combines:

- A linear trend estimated with `sklearn.linear_model.LinearRegression`.
- Median seasonal indices by calendar month.

A log-linear trend specification was also tested. It performed worse on the same holdout period, with 25.4% MAPE versus 19.0% MAPE for the selected linear model.

### 5.2 Validation

| Item                  | Value                                 |
|-----------------------|---------------------------------------|
| Training period       | January 2014–June 2017                |
| Training observations | 42 monthly observations               |
| Holdout period        | July 2017–December 2017               |
| Holdout observations  | 6 monthly observations                |
| Validation metric     | Mean Absolute Percentage Error (MAPE) |
| Validation result     | 19.0% MAPE                            |

Validation performance varies across months. July and September have low error, while August, October, and December have materially larger error. The forecast is most appropriate for directional planning and scenario discussion.

### 5.3 Six-month planning range

After validation, the model is refit using all 48 months and forecasts January–June 2018.

| Scenario    | Six-month sales forecast |
|-------------|--------------------------|
| Pessimistic | Approximately \$247K     |
| Base        | Approximately \$305K     |
| Optimistic  | Approximately \$363K     |

The scenario band uses the validated 19.0% MAPE as its width. It is a validation-error-based planning range, not a formal prediction interval.

## 6. Management Action Plan

| Priority | Action                                 | Business rationale                                                                 | Success measure                                          |
|----------|----------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------|
| P1       | Fix loss-making sub-categories         | Tables, Bookcases, and Supplies have negative aggregate profit                     | Loss dollars; aggregate profit; loss-making-line rate    |
| P1       | Establish discount guardrails          | Average transaction-line profit becomes negative at the observed 30% discount tier | Profit margin and loss-making-line rate by discount tier |
| P1       | Audit Philadelphia profitability       | High sales coexist with negative profit                                            | City margin; discount rate; loss dollars; product mix    |
| P2       | Address October Consumer dip           | Consumer order count and AOV both decline                                          | Consumer orders; AOV; October sales; October margin      |
| P2       | Redesign November promotions           | Strong sales occur alongside below-typical margins                                 | Incremental profit; margin; retained sales volume        |
| P3       | Grow Central through profitable volume | Central has the lowest aggregate regional profit                                   | Orders; AOV; margin; total profit                        |

These actions are based on descriptive analysis and an associational regression. They should be piloted, measured, and compared with an appropriate baseline before full rollout.

## 7. Dashboard Guide

| Dashboard page                   | Decision purpose                                                                                                             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Overview                         | Monitor headline KPIs, category/region profitability, and seasonality                                                        |
| Forecast                         | Review validation, forecast horizon, base forecast, and scenarios                                                            |
| Profitability                    | Examine discount tiers, negative-profit sub-categories, the regression, and the Price-Volume-Mix profit-change decomposition |
| Customers & Geography            | Compare segments, regions, city profitability, and monthly margin                                                            |
| Management Action Plan           | Prioritize findings into actions, tests, and measurable success metrics                                                      |
| Customer Profitability Deep Dive | Identify high-sales/low-profit customers and loss-making customer detail                                                     |

## 8. Limitations

- Sample Superstore is a demonstration dataset, not operational data for a real retailer.
- The Profit field does not disclose COGS, freight expense, fulfilment cost, promotion cost, marketing spend, or inventory cost.

- The discount regression is associational rather than causal.
- The October analysis identifies the segment and mechanics of the sales decline, not its external cause.
- The forecasting series has only 48 monthly observations; monthly seasonal estimates rely on limited repetitions.
- The forecast does not incorporate planned promotions, price changes, inventory constraints, competitor activity, or macroeconomic information.

## 9. Data Source

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Kaggle — Sample Superstore dataset: [vivek468/superstore-dataset-final](#).

For reproducibility, download the source CSV directly from Kaggle and place it in the repository data/ directory. Do not redistribute the raw data unless the dataset license permits it.